



UNIVERSITY
OF APPLIED SCIENCES
UPPER AUSTRIA

Game Engine Architectures

Fachhochschule Oberösterreich

Agenda

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1. Overview & Motivation
 2. Basics
 3. Object-Oriented Programming
 4. Game Engine Design
 5. Game Loop and Physics
 6. Game Engine Components
 7. Common Design Patterns
 8. Game Evaluation



Introduction: Overview & Motivation

Overview

The Computer Scientist's Mind

- A multi-level architecture
- Users only see the tip of the iceberg



Overview

The Computer Scientist's Mind

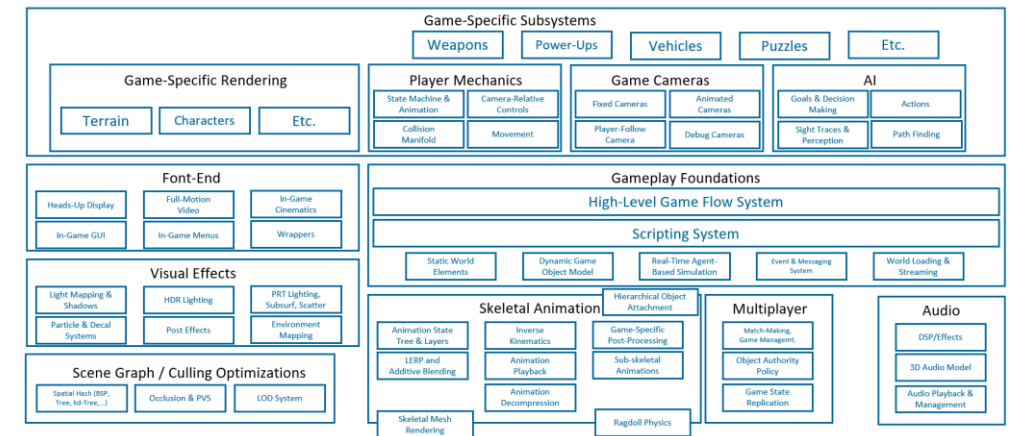
- Fully rendered 3D game scene with interactive components
- Beneath lies a 3D game engine



Overview

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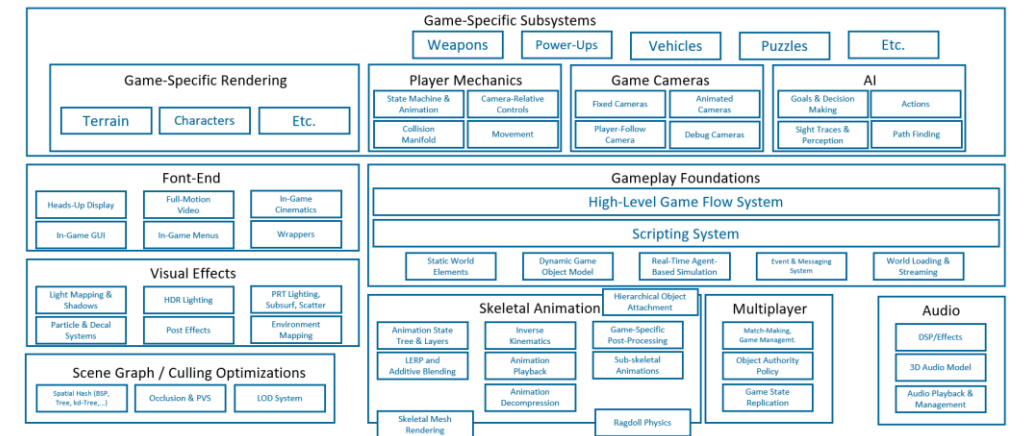
- The game engine is a software framework that separates between core components
- Graphics, game mechanics, physics/collisions, front-end, visual effects, scene graphs, etc.
- Components communicate and act based on code



Overview

The Computer Scientist's Mind

- The code is nothing else than subsequent instructions in a given programming language
- For a computer scientist, the language doesn't matter, because the concepts stay the same

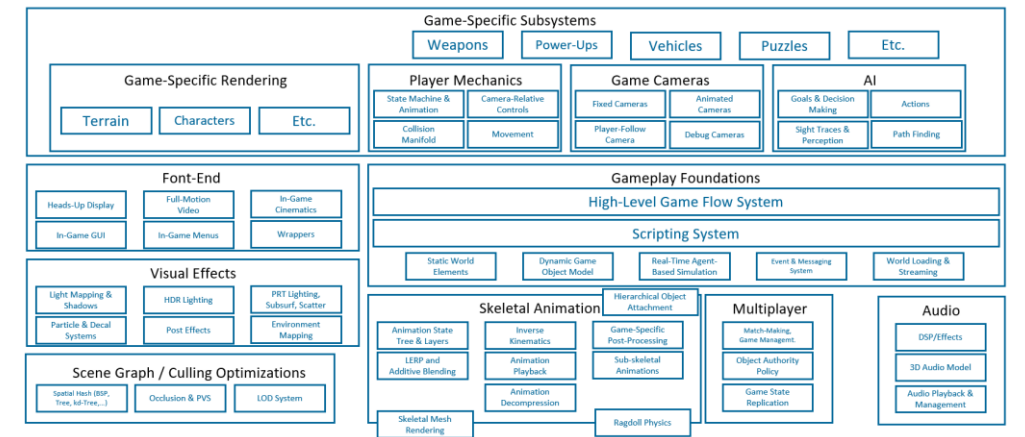


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// Called to bind functionality to input
void APlayerPawn::SetupPlayerInputComponent(UInputComponent* PlayerInputComponent)
{
    Super::SetupPlayerInputComponent(PlayerInputComponent);
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Overview

The Computer Scientist's Mind

- The programs themselves are translated into machine code
- The machine code is the atomic language of a particular computer
- Computers are both complex and entirely simple



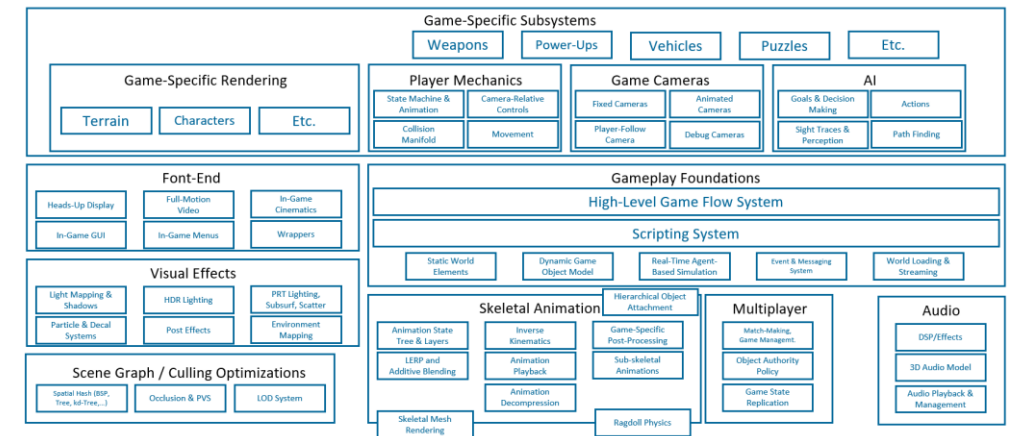
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0000000C: 2F 6C 61 6E 62 2F 4F 62 16 65 63 74 01 00 10 6A : /lang/Object@ j
000000D0: 61 76 61 2F 6C 61 6E 67 12F 53 79 73 74 65 6D 01 : ava/lang/System@
000000E0: 00 03 6F 75 74 01 00 15 14C 6A 61 76 61 2F 69 6F : vout@ java/io
000000F0: 2F 50 72 69 6E 74 53 74 172 65 61 6D 3B 01 00 13 : /PrintStream@ !!
00000100: 6A 61 76 61 2F 69 6F 2F 150 72 69 6E 74 53 74 72 : java/io/PrintStr
00000110: 65 61 6D 01 00 07 70 72 169 6E 74 6C 6E 01 00 15 : eam@ println@ $
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Overview

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- The programs themselves are translated into machine code
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- Computers are both complex and entirely simple
- **Over the semester, we will learn about these concepts on hand of the Unreal Engine, starting from the most atomic principles.**



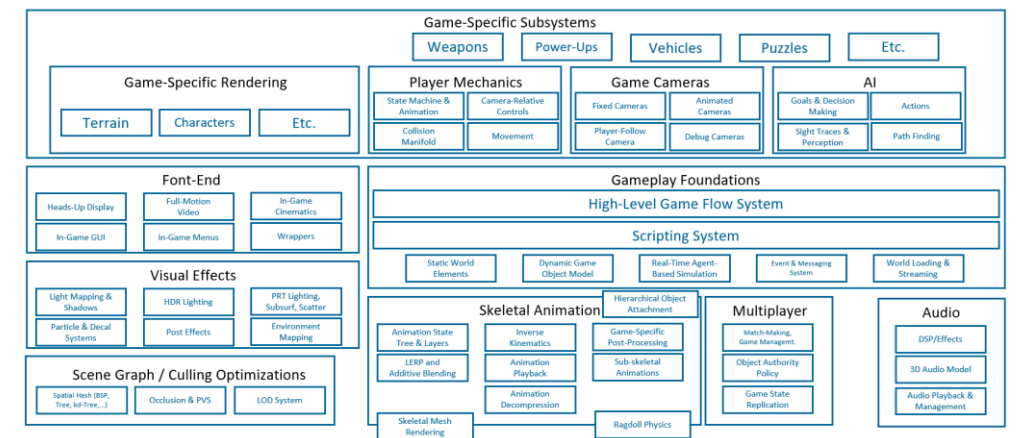
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- **Why? Unreal Engine is based on C++, which requires programming mastery!**



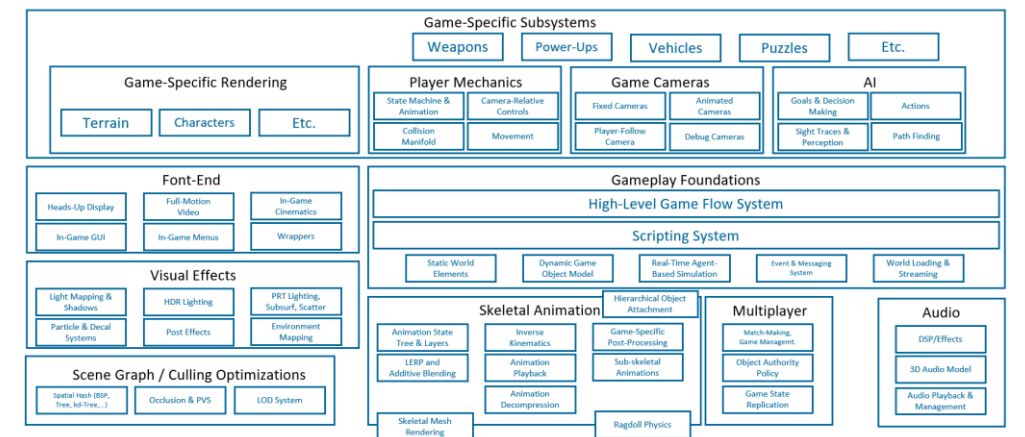
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Overview

The Computer Scientist's Mind

- The programs themselves are translated into machine code
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- Computers are both complex and entirely simple
- **Today, we will learn that “any program“ can be executed by a machine that provides only 1 instruction**



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Game Engine Architectures

Course Modalities



Overview

Course Modalities

→ Lectures

- 24 EH
- C++ and Unreal Engine
- Exam (tbd)

→ Exercises

- 16 EH
- Rui Kremismayr



Game Engine Architectures



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